

The Musical Intrigue of Pole-Zero Pairs

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All-pole filter banks have been used extensively in previous computer music analysis-resynthesis applications, but pole-zero filters are somewhat less common, except in the emulation of traditional analog filter types such as parametric equalizers, Chebyshev filters, etc.

For a small number of filter coefficients, the perceptually significant features of the spectral response of an instrument can be better resolved by including zeros as well as poles.

This paper will review the advantages of pole-zero (ARMA) filters over all-pole (AR) filters, and discuss the results of using several existing ARMA filter design methods for modeling the spectrum of a guitar body.

All-Pole filter banks have been used extensively in previous computer music analysis-resynthesis applications. All-pole filter banks offer considerable power in manipulating the spectral content of a signal. Also, methods are available to fit an all-pole filter bank to a given spectrum. However, using both poles and zeros in combination offers considerably more power than using either poles or zeros alone, and methods do exist for fitting an pole-zero filter bank to a given spectrum, although the reliability of these methods do not yet match the methods available for all-pole methods. Nonetheless, the additional control offered by poles and zeros in combination should be considered by the computer music system designer, especially when designing real-time systems, where processing power is still a very precious resource.

To review, an n-th order all-pole filter (3) has a transfer function given by

$$H(z) = G \frac{1}{B(z)} = \frac{G}{1 + b_1 z^{-1} + \dots + b_n z^{-n}}$$

and an n-th order rational transfer function (3) (with both poles and zeros) is given by

$$H(z) = \frac{A(z)}{B(z)} = \frac{a_0 + a_1 z^{-1} + \dots + a_n z^{-n}}{1 + b_1 z^{-1} + \dots + b_n z^{-n}}$$

Transfer functions have many implementations of course, with parallel and cascade forms being the most common (followed by lattices and waveguides). Typically, transfer functions higher than second order are not implemented in the direct form due to finite precision coefficient problems, except for fairly low Q filters. Instead, they are factored into first and second order sections and then either they are connected in series (cascade form decomposition) or their outputs are summed together in parallel (parallel form decomposition).

The partial fraction decomposition (3) given by

$$C + \sum_{i=1}^k \frac{a_{0i} + a_{1i} z^{-1}}{1 + b_{1i} z^{-1} + b_{2i} z^{-2}}$$

can implement any rational transfer function, but has a disadvantage of lacking independent parameters for the position of the zeros. The poles are independently controlled by each separate denominator but the zeros are determined by how the second order sections add together.

This may be easier to see by summing together two all-pole transfer functions with no zeros (only the constant gain terms A and C) and second order terms B(z) and D(z):

$$\frac{A}{B(z)} + \frac{C}{D(z)} = \frac{A D(z) + C B(z)}{B(z) D(z)}$$

In order to obtain a common denominator, the numerator $A D(z) + C B(z)$ is required which introduces zeros. While it looks at first like A and C are independent gain terms for each pole pair, they actually vary the position of the zeros! The parallel form is widely used, typically ignoring the effects of the zeros introduced. In practice, if the zeros cause problems with undesired cancellation, they are sometimes moved to reduce their undesired cancellation, but not to explicitly make use of the zeros in controlling spectral response (8,9). Quantization error in any coefficient, including the pole coefficients, can also cause the zeros to move.

The cascade form (3) has a transfer function of

$$a_0 \prod_{i=1}^k \frac{1 + a_{1i}z^{-1} + a_{2i}z^{-2}}{1 + b_{1i}z^{-1} + b_{2i}z^{-2}}$$

and is also known as series connection of second order sections. The i -th zero pair location is explicitly controlled by the a_{1i} , a_{2i} coefficient pair. The cascade form has disadvantages of course, such as ordering and pairing constraints (see below).

The standard parametric equalizer (figure 1), also known as the presence filter, illustrates many of the advantages of combining poles and zeros. First, the parametric equalizer offers independent control of frequency, bandwidth, and boost or cut amplitude. Second, it offers very localized control over spectral amplitude. Third, very high Q peaks and notches are possible.

A very simplified qualitative description of the parametric equalizer design is to put a pole and zero on the same radial line at $\pi/2$. The boost or cut amplitude is controlled by varying the radius of the poles and zeros; the equalizer will boost the amplitude if the pole is closer to the unit circle than the zero, and the equalizer will cut the amplitude (produce a notch) if the zero is closer to the unit circle. The closer to each other the pole and zero are, the more localized their effect is. Farther away from the boost/cut region, the pole and zero cancel each other out. The filter design equations are very simple, and well documented in a number of references (5,6,7).

Localized control of spectral magnitude is simple with this filter since outside of the boost/cut region, the parametric equalizer magnitude is unity. This result is hard to achieve with low order all-pole or all-zero realizations. A pole pair produces gain in one region, but falls off at 12 dB per octave, affecting the spectrum across the whole frequency range. So if a boost is desired at some low frequency band, loss at higher frequencies is unavoidable, and vice-versa. Pairing poles and zeros can circumvent this problem.

Very high Q peaks, notches, and spectral transitions can be produced with this filter type. It is easy to produce 40 to 50 dB peaks and deep notches with a second order parametric equalizer, while keeping the response flat away from the boost cut region.

One problem with low order all-pole modeling is passing high frequencies. As an example, five formants in the range of 0 to 5000 Hertz are often used to model speech vowels. Five complex poles produce a response falling off at $5 * 12 \text{ dB} = 60 \text{ dB}$ per octave above 5000 Hz, essentially eliminating high frequencies. In computer music applications where arbitrary excitations might be desired for such a filter bank, passing high frequencies is quite desirable. Pairing poles and zeros together allows easy tailoring of the high frequency response.

One intuitive suggestion to allow better high frequency transmission with high order all-pole filter banks is to mix back in some of the original signal. This technique works, of course, but looking at the transfer function of this operation we see that

$$1 + \frac{1}{B(z)} = \frac{B(z)}{B(z)} + \frac{1}{B(z)} = \frac{B(z) + 1}{B(z)}$$

which means that adding in the original signal to an all-pole system adds zeros! These zeros also decrease the Q of the poles, which suggests that a better approach is to explicitly exploit zeros to achieve desired goals.

While the cascade form decomposition for transfer functions makes using zeros easier, using zeros also makes the cascade form easier to use! By pairing each pole pair with the zero pair closest to it, the overall dynamic range of the filter cascade can be minimized, which improves signal to noise and overflow risk substantially. All-pole cascades require careful ordering in order to optimize signal to noise ratio (11). High Q Poles can amplify noise (quantization error) introduced by the attenuation of earlier low-frequency poles. In contrast, paired poles and zeros reduce or eliminate the need to order cascade sections, since the pole-zero pair only tends to affect a limited spectral region. Each zero should precede its associated pole (Direct Form I (3)). The pole will then boost the signal attenuated by the preceding zero, balancing the overall dynamic range. The peak frequency response magnitude (the L_∞ frequency response norm) of the overall second order section is known a priori from the Boost parameter, and the norms of the internal nodes of the individual second order section are easy to compute, simplifying scaling.

Analysis Methods

It seems that pole-zero filters have not been used much in analysis-resynthesis applications because this problem is still considered unsolved. Well-developed analysis methods are available to fit an all-pole filter to a given signal or response curve, such as Linear Predictive Coding (LPC) methods. The problem of pole-zero filter design, however, is not straightforward (1, 2,4,10).

Finding LPC coefficients is relatively easy because the prediction error performance surface is quadratic and hence has a unique global minimum. The LPC coefficients can be computed by simply solving a system of only linear equations. In contrast, pole-zero filter coefficients can not be computed by simply solving a system of only linear equations (2).

Iterative numerical techniques must then be employed to compute these coefficients. Since the equations are non-linear, the performance surface may have multiple local minima, thus these iterative techniques may yield sub-optimal solutions. Worse yet, the number of local minima can be infinite (2), making exhaustive search problematic (4).

Figure 2 shows the results of experiments to compare several pole-zero modelling methods. More specific details of this work are presented elsewhere (1,10), but to summarize, the goal was to model a guitar body impulse response with a low order (16th poles, 16th zeros) pole zero filter. The FFT of the guitar body impulse response was pre-processed to enhance perceptually significant features. The low-order filter was then fit to this pre-processed data.

Four methods were compared; Hankel Method, Least Squares Modified Yule-Walker Equations, an Auto-Regressive method (Yule-Walker equations, using the Levinson-Durbin recursion), and Steiglitz-McBride (Jackson). The equation error methods (LSMYWE) tend to underestimate the Q of poles close to the unit circle, but seem to fit the overall shape of the desired spectrum fairly well. The Hankel Norm method (2) works surprisingly well, yielding a better fit for the high Q low frequency poles, and actually producing a much better least square error than the other methods.

Conclusions

Poles and zeros in combination offer substantial benefits to the computer music designer, and methods to analytically fit pole-zero filters do exist, although this is still an active area of research.

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Figure 1 - Parametric Equalizer

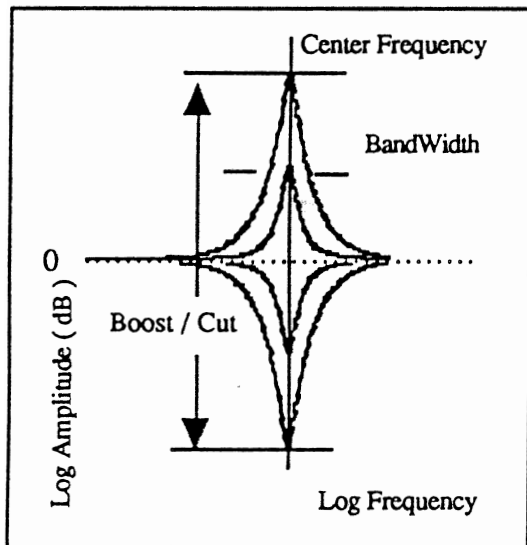
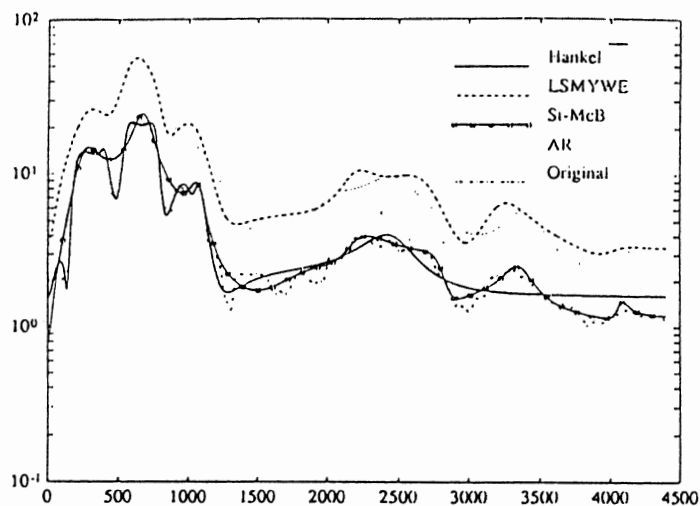


Figure 2. Pole/Zero Analysis Results (perceptually smoothed)



Method	ϵ
Hankel	4.3280 e-05
LSMYWE	2.6743 e-03
AR	2.6283 e-03
SMM	3.8258 e-04

$$\frac{1}{N} \sum_{n=1}^N [\tilde{h}(n) - \tilde{h}_d(n) - \bar{e}]^2$$